

Hidden Voices: Exploring Gender Disguise and Literary Parallels in Historical Authorship

A Machine Learning Approach to Classifying Author Gender from Biographical and Inferred Data

Israel Davidson

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1. Introduction and hypothesis

Introduction

Throughout history, the roles and abilities of women have often been underestimated, leading to societal restrictions that prevented them from fully participating in many fields. Women's contributions were frequently dismissed, and many were persecuted for engaging in activities considered outside the scope of traditional female roles. For example, the mere act of a woman being literate—knowing how to read, write, and perform basic mathematical operations—was once deemed suspicious and could even lead to accusations of witchcraft. This societal limitation contributed to a phenomenon where many women had to disguise their true identities in order to access opportunities reserved for men.

Throughout history, there are many stories of women who assumed male personas to participate in male-dominated fields. Frances Clayton, for instance, was an American woman who disguised herself as a man to serve in the Union Army during the Civil War. Similarly, Mary Ann Evans, one of the most influential English novelists of the 19th century, wrote under the male pseudonym George Eliot. These instances reflect the lengths to which women had to go to gain recognition and the ability to pursue their talents in areas traditionally reserved for men.

Hypothesis

Given the historical context where women often had to conceal their identities to participate in various professions and causes, it is hypothesized that books written by women will exhibit similarities to books written by men whose gender is only inferred from their names. This assumption is based on the belief that, despite the challenges faced by women, their writing styles

and content would align with the broader literary trends of their time, as women writers often adopted male pen names to avoid bias or limited opportunities.

2. Data and methods

```
import os
import pandas as pd
from sklearn.feature_extraction.text import TfidfVectorizer
from sklearn.decomposition import PCA, TruncatedSVD
from sklearn.preprocessing import Normalizer
from umap import UMAP
import seaborn as sns
import matplotlib.pyplot as plt
import numpy as np
from sklearn.cluster import KMeans
from imblearn.over_sampling import SMOTE
from sklearn.preprocessing import LabelEncoder

from sklearn.model_selection import cross_validate
from sklearn.neighbors import KNeighborsClassifier
from sklearn.linear_model import LogisticRegression
from sklearn.tree import DecisionTreeClassifier
from sklearn.ensemble import RandomForestClassifier
from sklearn.naive_bayes import MultinomialNB
from sklearn.preprocessing import StandardScaler
from sklearn.model_selection import GridSearchCV, cross_validate

from sklearn.metrics import accuracy_score,
f1_score, roc_auc_score, classification_report

import altair as alt
```

```
C:\Users\izzyd\OneDrive\Documents\School\Cornell\Women-in-
Writing\Women\.venv\Lib\site-packages\tqdm\auto.py:21: TqdmWarning: IProgress
not found. Please update jupyter and ipywidgets. See https://ipywidgets.
readthedocs.io/en/stable/user_install.html
  from .autonotebook import tqdm as notebook_tqdm
```

```
# Load the metadata
metadata = pd.read_csv(
    'corpus_data.tsv',
    sep='\t',
    low_memory=False
).set_index('source_id')
```

```
# Glance length of data
print(len(metadata))
```

```
1540
```

```
metadata=metadata[['gender', 'gender_guess', 'occupation']]
```

Field definitions and distributional stats

Most of the metadata fields are self-explanatory, but here are some details. Note that not every field in the metadata is described below.

- `source_id`: This is the name of the file corresponding to the volume. You can use it to match metadata records to full-text documents. Note that the corpus includes a nontrivial number of multivolume works. These volumes have `source_ids` like `eaf086v1` or `Wright2-1720v2`.
- `gender`: Author gender. M, F, or NaN (= unknown).
- `gender_guess`: Was the author gender assignment determined by biographical research (0) or by guessing on the basis of the author's name (1)?
- `occupation`: The author's primary employment identification. Recall that the US in the nineteenth century didn't always have a large market for novels, so many of the authors in the corpus made their living by other means.

```
metadata.head()
```

	gender	gender_guess	occupation
source_id			
eaf001	M	0.0	Arts
eaf002	F	0.0	Education
eaf003	F	0.0	Education
eaf004	F	0.0	Education
eaf005	M	0.0	Church